R16

Code No: 135CX

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, November/December - 2018
PRINCIPLES OF ELECTRONIC COMMUNICATIONS
(Common to CE, EEE, CSE, EIE, IT)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A

(25 Marks)

1.a)	Draw the block diagram of a communication system.	[2]
b)	Explain how baseband transmission is different from broad band transmission?	[3]
c)	Define amplitude modulation.	[2]
d)	Write the advantages of digital modulation schemes.	[3]
e)	Explain the operation of a bridge.	[2]
f)	Explain the operation of a NIC.	[3]
g)	Define numerical aperture.	[2]
h)	Draw the block diagram representing the basic elements of a fiber optic communication system.	[3]
i)	What is handoff?	[2]
j)	Discuss common Zigbee network topologies.	[3]

PART - B

(50 Marks)

2. Write short notes on:
a) Gain
b) Attenuation
c) Decibels. [10]

OR

3. Why modulation is used in communication circuits? Discuss. [10]

4.a) Draw the block diagram of PCM system and explain.
b) Define PWM and explain the concept of PWM using relevant waveforms. [5+5]

OR

5.a) Explain the basic principle of frequency modulation. Also explain a method to generate FM signal.
b) Explain a method to demodulate AM signal. [5+5]

6.a) Explain the operation of a paging system using a block diagram.
b) Classify computer networks and explain. [5+5]

OR

7. Discuss the topology, encoding and transmission media used in Ethernet LANs. [10]

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8.a) Explain the operation of telemetry, command and control subsystems of satellite.
b) Explain the operation of a GPS receiver using a block diagram. [5+5]

OR

9.a) Draw the general block diagram of an earth station and explain.
b) Explain single mode step index and multimode graded index fiber optic cables. [5+5]

10.a) Draw the block diagram of a RFID tag and explain.
b) Discuss GSM system in detail. [5+5]

OR

11.a) Discuss the basic forms of UWB wireless and also different types of UWB modulations.
b) Discuss WCDMA in detail. [5+5]

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